

Gaozheng Li

Email: ligzh2023@shanghaitech.edu.cn | Tel: +86 156-0179-2085

Education

ShanghaiTech University

B.Eng. in Computer Science and Technology

Cumulative GPA: 3.76/4.0 | TOEFL: 5.5/6 (equiv. 110/120)

Selected Coursework: Deep Learning (A+), Advanced Mathematics/Calculus (A+), Data Structures and Algorithms (A), Computer Architecture (A), Linear Algebra, Discrete Mathematics, Probability and Statistics

Shanghai, China

Sep.2023-Jun.2027(expected)

University of California, Berkeley

College of Engineering | Exchange Student

Cumulative GPA: 4.0/4.0

Berkeley, CA

Jan.2026–May 2026

Publications

[1] Sihan Ren¹, **Gaozheng Li**¹, Yuanshang Quan, Yiming Qin, Fuyi Yang, Chang Liu, Lan Xu, Minye Wu, “**ReSCUE: Re-Translation with Sentence Commitment for Unsegmented Long-Form Simultaneous Sign Language Translation**”, *NeurIPS 2026*.

[2] Chengyang Li, Hongdi Yang, **Gaozheng Li**, Zhenxuan Wu, Jingya Wang, Jingyi Yu, Zhuo Su, Lan Xu, “**SMGDiff: Soccer Motion Generation using Diffusion Probabilistic Models**”, *ICCV 2025*, <https://arxiv.org/abs/2411.16216>.

Research Experience

DreamerOptic: Motion-Supervised World Models for Reinforcement Learning

Jun.2026-Present

Supervised by Prof. Cathy Wu | Massachusetts Institute of Technology

- Developed DreamerOptic, extending DreamerV3 with action-conditioned optical-flow supervision to study whether explicit motion representations improve control-relevant world modeling.
- Explored compact motion representations for recurrent world-model dynamics and designed controlled experiments across architectures and conditioning strategies.
- Improved average policy performance by 11% across six challenging DeepMind Control Suite tasks over the DreamerV3 baseline.
- Found that substantially reducing optical-flow prediction error did not consistently translate into higher policy returns, motivating further investigation into which predictive representations are actually useful for downstream control.

ReSCUE: Unsegmented Long-Form Simultaneous Sign Language Translation

Apr.2025-Nov.2025

Supervised by Prof. Jingyi Yu & Lan Xu | ShanghaiTech University

- Led the development of ReSCUE, a streaming SLT framework that extends sign language translation from offline, pre-segmented sentences to continuous, unsegmented long-form video.
- Designed inference-aware training, stabilized re-translation, and sentence commitment to handle partial observations, output instability, ambiguous sentence boundaries, and growing inference context.
- Built the large-scale multimodal training and evaluation pipeline, including distributed LLM fine-tuning and long-form data construction; achieved strong quality–latency trade-offs and near-oracle translation quality on long-form benchmarks.

SMGDiff: Controllable Soccer Motion Generation with Diffusion Models

Sep.2024-Nov.2024

Supervised by Prof. Lan Xu | ShanghaiTech University

- Identified unrealistic ball-foot contact failures in generated soccer motions and developed a contact-guidance mechanism that infers likely contacts from ball acceleration and enforces physically consistent interactions during diffusion sampling.
- Empirically calibrated the contact threshold from motion data and incorporated it into a contact loss, reducing unrealistic ball drift and improving player–ball interaction.
- Contributed to SMGDiff’s real-time controllable motion-generation pipeline and the Soccer-X motion-capture dataset, containing 1.08M frames from 30 players.

Honors & Awards

Outstanding Student (Top 10%) | ShanghaiTech University

Oct.2025

Outstanding Student (Top 10%) | ShanghaiTech University

Oct.2024

Outstanding Member | ShanghaiTech Student Union

Aug.2024

Professional Skills

Programming: Python (PyTorch), C/C++, MATLAB

ML/Research: PyTorch, JAX, Hugging Face, Weights & Biases

Tools: Linux/Ubuntu, Git, L^AT_EX, Unity3D

Activities

Student Delegate | World Laureates Forum

2023

Campus Champion | RoboMaster (ShanghaiTech)

2024

Student Officer | ShanghaiTech Student Union (Arts & Sports Department)

2023–2024